

# Moment of Untruth: Dealing with Negative Queries in Video Moment Retrieval

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## Abstract

*Video Moment Retrieval is a common task to evaluate the performance of visual-language models—it involves localising start and end times of moments in videos from query sentences. The current task formulation assumes that the queried moment is present in the video, resulting in false positive moment predictions when irrelevant query sentences are provided.*

*In this paper we propose the task of Negative-Aware Video Moment Retrieval (NA-VMR), which considers both moment retrieval accuracy and negative query rejection accuracy. We make the distinction between In-Domain and Out-of-Domain negative queries and provide new evaluation benchmarks for two popular video moment retrieval datasets: QVHighlights and Charades-STA. We analyse the ability of current SOTA video moment retrieval approaches to adapt to Negative-Aware Video Moment Retrieval and propose UniVTG-NA, an adaptation of UniVTG designed to tackle NA-VMR. UniVTG-NA achieves high negative rejection accuracy (avg. 98.4%) scores while retaining moment retrieval scores to within 3.87% Recall@1. Dataset splits are available at <https://github.com/keflanigan/MomentofUntruth>*

## 1. Introduction

With the ever-increasing amount of video data that is accessible to the public through streaming websites, the ability to quickly search through this data is attaining an increased importance. Not only is it necessary to search for relevant videos, it is also desirable to search through the videos themselves. The video moment retrieval task addresses this by searching for relevant moments within videos using text queries as input. Currently, video moment retrieval models focus only on producing accurate start and end times for text queries, under the assumption that the moment *always exists* within the video. Video moment retrieval datasets are composed of video-sentence pairs which all have a direct correspondence through labelled start and end times. However, this raises the question:

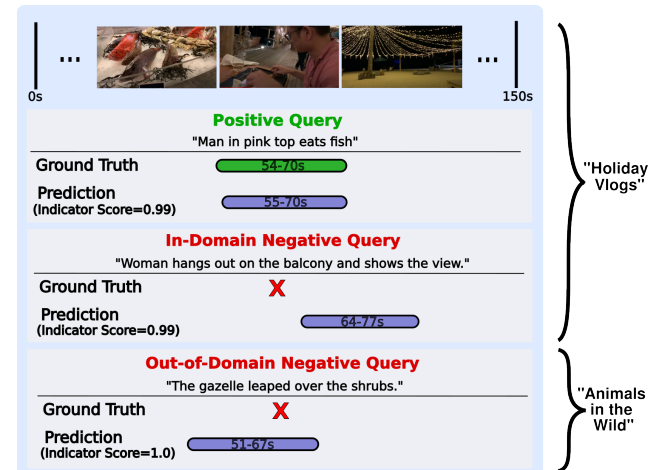


Figure 1. Video moment retrieval models are designed to predict start and end times in a video given a query sentence. Regardless of whether the text query is positive (exists in the video), in-domain negative (from the same domain but not present) or out-of-domain negative (from an entirely different scenario), current models such as UniVTG still produce a start-end time prediction.

*How do models perform with irrelevant textual queries for a given video?*

For example, suppose you have a video of a person eating dinner and then you ask the model to localise the sentence "the gazelle leaped over the shrubs". Should a video moment retrieval model return a start and end time for this sentence even though it clearly doesn't correspond with anything in the video? Current models will always provide a start and end time prediction regardless of the relevancy of the sentence, which can lead to hallucination (see Figure 1). Whilst there has been a great amount of progress in improving these models, they do not handle sentences which are irrelevant to the video, known as negative queries. We believe that if these models are to be robust and useful, they should be able to indicate when an input text query is irrelevant to the given video. We believe this task of negative rejection is a crucial aspect to ensure trustworthy and explainable AI models.

In this paper, we demonstrate, quantitatively and qual-

itatively, that current methods are not robust to negative queries and are not designed to distinguish between positive and negative queries. We do this by testing a combination of both in-domain, i.e. queries from the same dataset not relevant to the query video, and out-of-domain negatives, i.e. queries from a different scenario. We highlight examples of In-Domain (ID) and Out-Of-Domain (OOD) negatives within Figure 1 with example failure cases.

To combat this, we propose a method to reject negative queries while maintaining video moment retrieval performance. Our method uses an additional head which is explicitly trained to predict whether the sentence is relevant. We train the model with both ID and OOD negatives, showing both are necessary.

Overall, our contributions are as follows: (i) To the best of our knowledge, we conduct the first analysis of the robustness of SOTA video moment retrieval models to negative queries, while making the important distinction between in-domain and out-of-domain negative queries. (ii) We outline a method for sampling in-domain and out-of-domain negative queries for use during training and evaluation and provide new evaluation benchmarks for two popular Video Moment Retrieval datasets: QVHighlights [21] and Charades-STA [12]. (iii) We propose a new framework which enables existing video moment retrieval methods to train for Negative-Aware Video Moment Retrieval. (iv) We showcase strong results on negative rejection while retaining high moment retrieval performance from a model which has been adapted under our framework.

## 2. Related Work

**Moment Retrieval.** The task of video moment retrieval was first introduced in [1, 12], with the aim of expanding the action localisation task [34, 38] to the open vocabulary setting. Moment retrieval methods require cross-domain interactions between video and text in order to determine the correspondences between them. Approaches have historically been divided largely between proposal-based [1, 2, 4, 12, 13, 19, 25–27, 36, 40, 43, 49] and proposal-free methods [6–9, 15, 24, 30, 33, 37, 45, 47, 48, 50].

Recent works have been designed to jointly perform moment retrieval and highlight detection [14, 21], being trained with both **moment start/end times** and **saliency scores** which are the ground truths for the highlight detection task. Certain datasets such as QVHighlights [21] contain both human-annotated moments and saliency scores. Datasets which do not have human-annotated saliency scores instead use pseudo-saliency scores from the moment start/end times, with a non-zero score within ground truth moments.

**Moment Retrieval Methods.** Moment-DETR [21] is one such method that jointly trains on highlight detection and has served as a base for many recent moment retrieval meth-

ods [18, 20, 23, 28, 29, 31, 32, 39, 42, 44]. This approach takes inspiration from DETection TRansformer (DETR) [3] methods in object detection, viewing moment retrieval as a direct set prediction problem, generating moment candidates and associated scores with which to rank them in an end-to-end manner. It concatenates text and video features as input and passes them through a transformer encoder-decoder. Trainable positional embeddings known as *moment queries* are inputted to the decoder to generate candidate moment predictions. It has separate prediction heads for the moment span; the score of each proposed moment; and the saliency scores. While Moment-DETR uses foreground/background labels during training to supervise the moment score predictions, these are not designed to be used during evaluation for determining positive/negative queries. Some approaches based on this framework have included the addition of extra prior information to initialise the queries [18] or the addition of an extra modality [29]. Others have focused on improving the video-text interactions in the model [31, 32].

UniVTG [23] jointly trains for three tasks: moment retrieval, highlight detection, and video summarisation from datasets which do not contain a training signal for all three. It utilises a large pre-training scheme and alters the Moment-DETR architecture by removing the decoder and estimating an indicator score and predicted span for every feature clip in the video. This replaces the moment queries used within Moment-DETR. Furthermore, saliency scores are produced purely through feed-forward layers and attentive pooling over text features, without being passed through the transformer encoder.

QD-DETR [32] alters the encoder to contain cross-attention layers to ensure that the visual features are being properly attended to by the text features. It makes use of negative queries during training, shuffling video-sentence pairs across the dataset. This forces the saliency score predictions to be more indicative of the relevance of the sentence to the video clips. However, saliency score predictions are still not used for the video moment retrieval task.

CG-DETR [31] employs a clip-word correlation learner and uses dummy tokens concatenated with the text query tokens. These dummy tokens are designed to attend to sections of the video that are not represented by the query, thus essentially representing the query-excluded meaning. This helps to prevent irrelevant video clips from being represented by the text query.

Despite progress in moment retrieval performance, *all methods assume queries at inference are always positive*. Models are not evaluated with an input text query that is irrelevant to the video. Additionally, models are not equipped to handle such negative queries.

**Video Corpus Moment Retrieval.** Video Corpus Moment Retrieval (VCMR) [10, 17, 22, 46] is another related task. This expands the search from a single video to a corpus of

videos. While this returns only a single video segment from a set of videos, essentially serving as a rejection of the other videos, VCMR works under the assumption that the query is present in exactly one video in the corpus. There is no active scheme to determine negatives. Currently, VCMR methods cannot determine if a query is *not* present in any videos in the corpus, similar to the current state of video moment retrieval. Therefore this task formulation also does not allow for negative rejection in video moment retrieval.

**Negative Rejection in Other Tasks.** Modelling uncertainty of predictions, and discarding decisions with low certainty has been integrated into classification tasks [11, 16, 35, 41]. Our approach differs from these because the model is actively predicting whether the query is present in the video, rather than stating that it is unsure about the prediction.

Recently, the topic of negative rejection has been highlighted [5] in LLMs with Retrieval Augmented Generation, whereby information is extracted from retrieved documents to aid in responding to input queries. Negative rejection is crucial in cases where the required information is not present in the retrieved documents, as otherwise the hallucination of incorrect information can occur. It has been shown that currently these LLMs are not robust to negative rejection [5]. Our findings for the video moment retrieval task parallels this study, where current models are hallucinating moments in videos when negative queries are provided. Just as it will be important to deal with this issue in LLMs in order to improve reliability, it will also be important to achieve negative rejection in video moment retrieval.

### 3. Method

In this Section, we first present details of the standard Video Moment Retrieval Task and how Negative-Aware Video Moment Retrieval differs in Sec. 3.1, before defining types of negative queries in Sec. 3.2 and how they can be collected in Sec. 3.3. Lastly, we detail how current Video Moment Retrieval models can be trained for Negative-Aware Video Moment Retrieval in Sec. 3.4.

#### 3.1. Negative-Aware Video Moment Retrieval

We first present the Video Moment Retrieval task as defined in the literature. Formally, for each video  $V_i$  within a corpus, there exists a set of query sentences  $q_{i,j}$  with corresponding moments given as start  $t_{i,j}^s$  and end  $t_{i,j}^e$  times. We collectively describe this as the set of queries for video  $V_i$ :  $Q_i = \{(q_{i,j}, t_{i,j}^s, t_{i,j}^e)\}$ . During training, models learn to predict the start/end times of a moment given the corresponding query sentence for the  $i$ th video.

At inference time, methods are evaluated on their ability to correctly localise the query sentence  $q_{i,j}$  which is always assumed to be contained within video  $V_i$ . Therefore, methods rank and select the highest proposal/predicted mo-

ment  $(\tilde{t}_{i,j}^s, \tilde{t}_{i,j}^e)$  and compare this to the ground truth moment  $(t_{i,j}^s, t_{i,j}^e)$  directly during evaluation. By doing so, all Video Moment Retrieval methods make the assumption that all query sentences  $q_{i,j}$ , positive or negative, are relevant and contained within a video  $V_i$ .

In this work, we propose Negative-Aware Video Moment Retrieval (NA-VMR) in which models should reject negative query sentences which are not contained within the video and return a moment span only for positive query sentences that are contained within the video. The model therefore predicts the start/end times as before as well as whether to accept or reject the query. Formally, the model will output a tuple  $(\tilde{y}, \tilde{t}^s, \tilde{t}^e)$  containing the prediction score,  $\tilde{y}$ , and the predicted start/end times  $\tilde{t}^s$  and  $\tilde{t}^e$ . In the case of a negative prediction  $\tilde{y} = 0$ , then  $\tilde{t}^s$  and  $\tilde{t}^e$  are considered invalid and rejected. For a positive prediction score ( $\tilde{y} = 1$ ) the predicted start/end times are considered valid and compared to the ground truth moments as normal.

For both training and evaluation both positive and negative queries need to be utilised. We define positive queries for video  $V_i$  as a tuple of  $Q_i^+ = \{(y_{i,j} = 1, q_{i,j}, t_{i,j}^s, t_{i,j}^e)\}$ , and negative queries  $Q_i^- = \{(y_{i,k} = 0, q_{i,k})\}$ . Next, we provide more detail regarding the negative queries.

#### 3.2. In-Domain & Out-of-Domain Negatives

We choose to divide negative queries into two categories, namely **In-Domain (ID)** and **Out-of-Domain (OOD)**, which represent queries from a similar context to the video and queries from a different context. The distinction is related to the plausibility of the query being present in the video. Both sets of negatives are important to consider when examining the behaviour of moment retrieval models. ID negatives allow for the inspection of a model’s ability to differentiate specific details in videos, while OOD negatives enable the inspection of a model’s ability to recognise that a query is entirely irrelevant to the scenario. Regardless of whether the negative is in-domain or out-of-domain, a Negative-Aware Video Moment Retrieval method should correctly recognise that there is no corresponding moment to retrieve from the video.

**In-Domain Negatives** are queries describing events which do not occur in a given video, but which feasibly could occur within a video from that domain or context. For example, in videos of a person carrying out actions in a kitchen, the sentence “the person opens the oven” would be plausible, and would be an ID negative if no oven was opened within this video.

**Out-of-Domain Negatives** are defined as queries which belong to an entirely different scenario to the selected video, and which are therefore extremely unlikely to be present within it. An example is the sentence “the player hits the ball across the net” for the above video of a person carrying out actions in a kitchen.

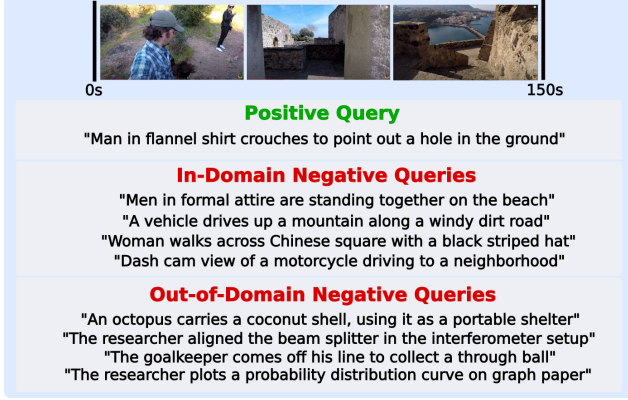


Figure 2. In-Domain and Out-of-Domain negative queries alongside a corresponding positive video-sentence pair.

### 3.3. Sampling Negative Queries

We sample ID and OOD negatives to ensure that these represent true negatives whilst still including a wide variety of queries. Examples of these are displayed in Figure 2.

**In-Domain Negatives** are produced by shuffling video-sentence pairs within the dataset, similar to the method used in [32], assigning a different video to each sentence. We choose not to generate new ID negatives to reduce the chance of false negatives in the test set, which occur when a moment described by that query sentence is present in the newly assigned video. We ensure a clean test set as follows: Firstly, cosine similarity scores are calculated for each sentence with every other sentence in the test set by using CLIP embeddings. Then, a pseudo-similarity score is calculated between a sentence and every video in the test set by determining the highest sentence-sentence similarity score for each video. For example, to get the similarity score between query sentence  $q_k$  and video  $V_i$ , you first calculate the cosine similarity between  $q_k \notin Q_i$  and each  $q_{i,j} \in Q_i$  and take the maximum of those similarity scores to be the sentence-video pseudo-similarity score. Each sentence is then assigned to a video whose video-sentence similarity score is in the lowest 50th percentile for that sentence.

**Out-of-Domain Negatives** are generated via a large language model (LLM). Scenarios are selected whose actions would be extremely unlikely to occur within the specific dataset. Within these scenarios, the LLM is prompted to generate sentences describing actions from a variety of subtopics within the scenario. For example, for a cooking dataset, the scenario of competitive sport might be chosen. Accordingly, the LLMs are prompted to generate sentences describing actions occurring in topics such as "football", "basketball", "tennis" and others. We select the scenarios based on the dataset to ensure that there is no overlap and generated sentences are not false negatives.

### 3.4. Modelling with Negative Queries

Many current Video Moment Retrieval methods are based on Moment-DETR [21]. We detail the general concept of these methods first before describing extensions towards Negative-Aware Video Moment Retrieval.

**Video Moment Retrieval Methods.** Current Moment-DETR-based methods utilise frozen video and text encoders, usually followed by a projection layer to match dimensionality. This produces video features  $\mathbf{V} = \{\mathbf{v}_c\}_{c=1}^{L_v}$  and text features  $\mathbf{Q} = \{\mathbf{q}_w\}_{w=1}^{L_q}$  where  $L_v$  is the number of video clips and  $L_q$  is the number of query sentence tokens. Methods typically pass these video and text features into a Transformer encoder to produce text-attended video tokens. It is common to use a Transformer decoder with  $M$  trainable position embeddings known as *moment queries* as input alongside the text-attended video tokens, thus producing  $M$  final video representations which may be used as input to the heads. Certain methods such as UniVTG [23] instead directly utilise the text-attended video tokens as input to the three predictions heads.

These methods use three prediction heads. Firstly the foreground matching head produces the indicator scores  $\tilde{f}_m$ , where  $m$  is the index of the candidate moment predictions. These are typically produced by a set of feed forward layers and an activation function on top of the  $M$  video representations and aim to predict the likelihood of the moment matching the query. Similarly, the moment boundaries  $\tilde{t}_m$  are predicted via another boundary prediction head, but with two outputs: either start/end time offsets or moment centre and width. There can also be a saliency head to predict the saliency scores  $\tilde{s}_c$ , where  $c$  is the clip index. The input to the saliency head is typically the output of the video encoder or earlier representations. The saliency scores are used only for the highlight detection task, which aims to detect text-guided highlights for a given video.

Whilst specific details of how these moment and saliency predictions are produced vary, the basic principles outlined above remain the same.

**Incorporating Negative Queries.** We propose to alter Moment-DETR-based methods as follows. The generated negative queries are used to train the model to differentiate between positive and negative video-sentence pairs, enabling Negative-Aware Video Moment Retrieval. As shown in Figure 3, we add a binary classification head on top of the base Video Moment Retrieval model which classifies queries as positive or negative. The indicator score and saliency score predictions are combined and passed into a recurrent (RNN) layer. Both are used, as while the indicator scores denote the likelihood of the moment matching the query, saliency scores from Figure 4 are much more discriminative between positive and negative. We employ an RNN to maintain temporal knowledge within the features



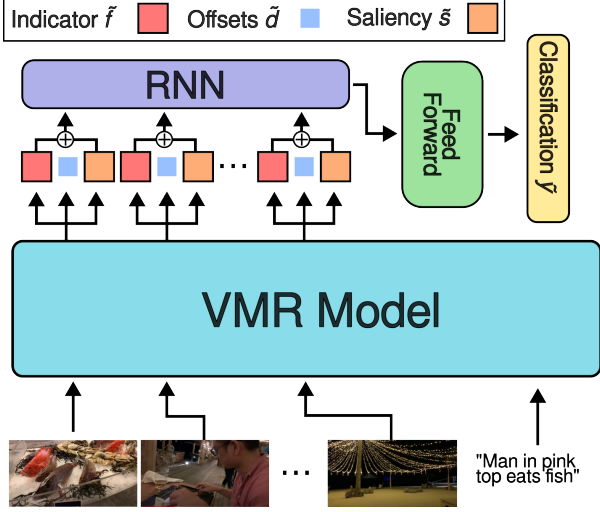


Figure 3. The classification head for NA-VMR is added to a Video Moment Retrieval model (UniVTG in this case) via summation of the indicator and saliency scores, which are then passed through a recurrent layer and a feed forward layer before producing a single value output for classification.

while handling variable video lengths in a lightweight manner. The RNN output at the final step is then passed into a feed-forward layer (MLP) and, finally, a sigmoid activation function to produce the final prediction score,  $\tilde{y}$ .

$$\tilde{y} = \sigma(\text{MLP}(\text{RNN}(\tilde{f} \oplus \tilde{s}))) \quad (1)$$

where, the  $\oplus$  operation in Equation 1 represents a generic combination. This can be a concatenation, learned combination, or in our case, a summation of the indicator and saliency scores. We denote models following this architecture *Negative-Aware* with NA as the suffix.

**Losses.** A binary cross entropy loss is applied to the classification head output, with positive queries taking a ground truth value of  $y = 1$  and negative queries taking a value of  $y = 0$ .  $\tilde{y}$  is the output prediction from the classification head and  $\lambda_p$  is the loss weighting. The loss is given as:

$$\mathcal{L}_p = \lambda_p(y \log \tilde{y} + (1 - y) \log(1 - \tilde{y})) \quad (2)$$

Typically, methods utilise a foreground matching head loss  $\mathcal{L}_f$  on the indicator scores, a boundary loss  $\mathcal{L}_b$  across the start/end times, and a saliency loss  $\mathcal{L}_s$  applied to the saliency scores. For positive queries, these losses can be applied as normal. However, losses on negative queries are adapted as follows:  $\mathcal{L}_b$  is set to 0 as there is no ground truth boundary to predict, while all ground truth values for the indicator and saliency scores are set to 0 for  $\mathcal{L}_f$  and  $\mathcal{L}_s$ . These loss functions may need to be adjusted if for example they involve contrastive losses, and as such are denoted  $\mathcal{L}_f^-$  and  $\mathcal{L}_s^-$  for negatives. This follows the understanding of their basic function; to lower  $\tilde{f}_i$  and  $\tilde{s}_i$  predictions for negative queries.

The original losses are combined with the proposed classification loss  $\mathcal{L}_p$ , which defines the losses for positives  $\mathcal{L}^+$  and negatives  $\mathcal{L}^-$ .

$$\mathcal{L}^+ = \mathcal{L}_f + \mathcal{L}_b + \mathcal{L}_s + \mathcal{L}_p \quad \mathcal{L}^- = \mathcal{L}_f^- + \mathcal{L}_s^- + \mathcal{L}_p \quad (3)$$

The total loss  $\mathcal{L}_{tot}$  is the weighted sum of the losses for positives  $\mathcal{L}^+$ , ID negatives  $\mathcal{L}_{ID}^-$ , and OOD negatives  $\mathcal{L}_{OOD}^-$ , where  $\lambda^+$ ,  $\lambda_{ID}^-$  and  $\lambda_{OOD}^-$  are the loss weights and  $\mathcal{L}_{ID}^-$  and  $\mathcal{L}_{OOD}^-$  are  $\mathcal{L}^-$  applied to the specific domains.

$$\mathcal{L}_{tot} = \lambda^+ \mathcal{L}^+ + \lambda_{ID}^- \mathcal{L}_{ID}^- + \lambda_{OOD}^- \mathcal{L}_{OOD}^- \quad (4)$$

While the exact losses may vary depending on the model architecture, we have described the basic principles of adjusting these losses to account for negative queries.

To summarise, we add an extra classification head on top of the indicator and saliency score predictions. This head classifies the query sentence as positive or negative. During training, the saliency and indicator scores are set to 0 for the negative queries and the losses are adjusted where necessary to ensure that the model is able to learn a strong signal to differentiate between positive and negative queries. Specific details for each model used may be found in the supp.

## 4. Experiments

We first present information on the evaluation protocol, giving details of the metrics and datasets used; implementation details; and baseline implementations.

**Metrics.** We use the standard moment retrieval metric of Recall@k with IoU@ $\theta$ , following the literature [21, 23, 31, 32], in specifying k=1 and  $\theta \in \{0.5, 0.7\}$ . We also introduce the metric of **Rejection Accuracy (RA)** which is the percentage of negative queries correctly rejected.

**Models.** We report results on UniVTG [23], CG-DETR [31], and QD-DETR [32], alongside their Negative-Aware adaptations, focusing primarily on UniVTG-NA.

**SVM Baseline.** We use SVMs as an alternative classifier for negative rejection. Saliency score outputs from the original model trained without negative query classification are passed through a Support Vector Machine (SVM). We use both ID and OOD negatives to train this SVM. For each video and query sentence pair in the training set, the top 3 predicted saliency score values across the video are found and averaged to provide a score of relatedness between query sentence and video. These values are passed into the SVM to train it to classify each query as either positive or negative. The trained model is then applied to test data to produce positive/negative classifications.

**Datasets.** We use the QVHighlights [21] and CharadesSTA [12] datasets, following [21]. QVHighlights consists of vlogs and news report videos sourced online, providing both human annotated moment start/end times and saliency

| Method           | $\tilde{f}$ | $\tilde{s}$ | QVHighlights  |               |                    |       | Charades-STA  |               |                    |       |
|------------------|-------------|-------------|---------------|---------------|--------------------|-------|---------------|---------------|--------------------|-------|
|                  |             |             | R1@0.5        | R1@0.7        | Rejection Acc. (%) |       | R1@0.5        | R1@0.7        | Rejection Acc. (%) |       |
|                  |             |             |               |               | ID                 | OOD   |               |               | ID                 | OOD   |
| UniVTG-Thr [23]  | ✓           | ×           | 67.23 (-0.12) | 52.52 (-0.13) | 8.97               | 6.52  | 60.19 (-0.03) | 38.55 (+0.00) | 0.56               | 0.24  |
| CG-DETR-Thr [31] | ✓           | ×           | 67.03 (-0.07) | 53.55 (+0.00) | 0.32               | 0.39  | 57.02 (-0.51) | 35.43 (-0.24) | 23.15              | 9.52  |
| QD-DETR-Thr [32] | ✓           | ×           | 61.94 (-0.06) | 46.19 (-0.07) | 0.39               | 0.06  | 58.92 (-0.19) | 36.64 (-0.11) | 4.19               | 4.57  |
| UniVTG-Thr [23]  | ×           | ✓           | 67.35 (+0.00) | 52.65 (+0.00) | 64.65              | 73.03 | 60.05 (-0.17) | 38.47 (-0.08) | 9.81               | 19.95 |
| CG-DETR-Thr [31] | ×           | ✓           | 66.58 (-0.52) | 53.23 (-0.32) | 82.00              | 86.45 | 56.85 (-0.68) | 35.43 (-0.24) | 10.86              | 6.29  |
| QD-DETR-Thr [32] | ×           | ✓           | 57.23 (-4.77) | 43.61 (-2.65) | 89.68              | 87.35 | 58.39 (-0.72) | 36.34 (-0.41) | 15.51              | 23.04 |

Table 1. Video Moment Retrieval results using a threshold on the indicator score ( $\tilde{f}$ ) and saliency score ( $\tilde{s}$ ). Thresholds are chosen by setting the 0.5th percentile on the positive training set. The differences to R1@ $\theta$  with no negative rejection are highlighted alongside.

scores for each video-sentence pair. We report results on the publicly available validation set of QVHighlights. Charades-STA is made up of home videos with scripted actions and provides just moment start/end times.

**Negative Queries.** Our OOD train and test sets have 7230 and 1550 text queries respectively, to match the numbers from the QVHighlights train and val sets. For the larger Charades-STA, the batch sizes are retained during training so some OOD negative queries are sampled twice during each epoch, but are assigned to random videos so produce a distinct training signal during each iteration. For the test set, OOD negative queries are assigned to multiple videos in order to match the larger number of positive queries in Charades-STA. ID negative sets always match the size of the positive sets.

**Implementation Details.** We use the same pre-trained model provided with UniVTG [23] as the base model for training UniVTG-NA on each dataset. For each model, RNN and feedforward layers all have a hidden dimension of 50. Batch sizes of 32 each are used for the positive, ID negative, and OOD negative queries during training. Loss weight values are found in the supp.

For OOD negative query generation, the broad topics of “animal behaviour”, “competitive sports”, “physics laboratory” and “mathematics class” are used. Within each scenario, more specific subtopics are used to generate queries. The subtopics are selected so as to produce a broad range of sentences across the scenarios. The LLMs used for generation are Claude-3-Opus and GPT-4o. Specific details of the prompts used and the subtopics are found in the supp.

#### 4.1. Shortcomings of Current Methods

In this section, we first explore how current methods that *have not been trained* to explicitly distinguish negative queries are able to perform at inference time on Negative-Aware Video Moment Retrieval. Specifically, we investigate current state of the art methods UniVTG [23], CG-DETR [31], and QD-DETR [32] on both QVHighlights [21] and Charades-STA [12].

Table 1 showcases an unsupervised approach, denoted

with a suffix -Thr, for each of the three methods using either indicator scores ( $\tilde{f}$ ) or saliency scores ( $\tilde{s}$ ) as a threshold for determining positive or negative queries. In both cases, thresholds were set to the lowest 0.5th percentile value of  $\tilde{f}$  or  $\tilde{s}$  in the training set of positive queries. This shows off-the-shelf performance from raw model outputs without requiring further modelling of the relationship between positive and negative query scores. The 0.5th percentile was chosen to set the classification boundary as the lower limit of the positive sentence scores, while mitigating the effect of outliers. The results show that methods are unable to distinguish between positives and negatives when the indicator score is used. Using saliency scores fares better for QVHighlights, though methods struggle on Charades-STA.

We analyse this further in Figure 4 which shows histograms of the methods across both datasets looking at the indicator score and the saliency score of positive and negative queries within the test sets. We note that the indicator score, which is the only score used for Video Moment Retrieval evaluation, is not separable and *methods treat positives and negatives the same way*. The saliency score provides more promise for separating positive and negative queries, but across all three methods there is still a considerable overlap between positives and negatives, especially on Charades-STA which doesn’t have ground truth saliency scores to train on, leading to poor rejection accuracy.

#### 4.2. Training with Negative Queries

In this section, we utilise our proposed negative queries at training time. In Table 2 we compare using an SVM to predict whether a query is positive or negative vs. our negative-aware (NA) methodology.

We find that each negative-aware method achieves strong Rejection Accuracy performance while retaining much of the Recall@1, IoU@k performance. Across both QVHighlights and Charades-STA the OOD Rejection Accuracy for each method is  $\sim 100\%$ , higher than the SVM. The ID Rejection Accuracy of each is over 90% for QVHighlights. The lower scores on Charades-STA likely stem from similar actions occurring in many videos within Charades-STA. While the SVM achieves higher ID Rejection Accuracy on

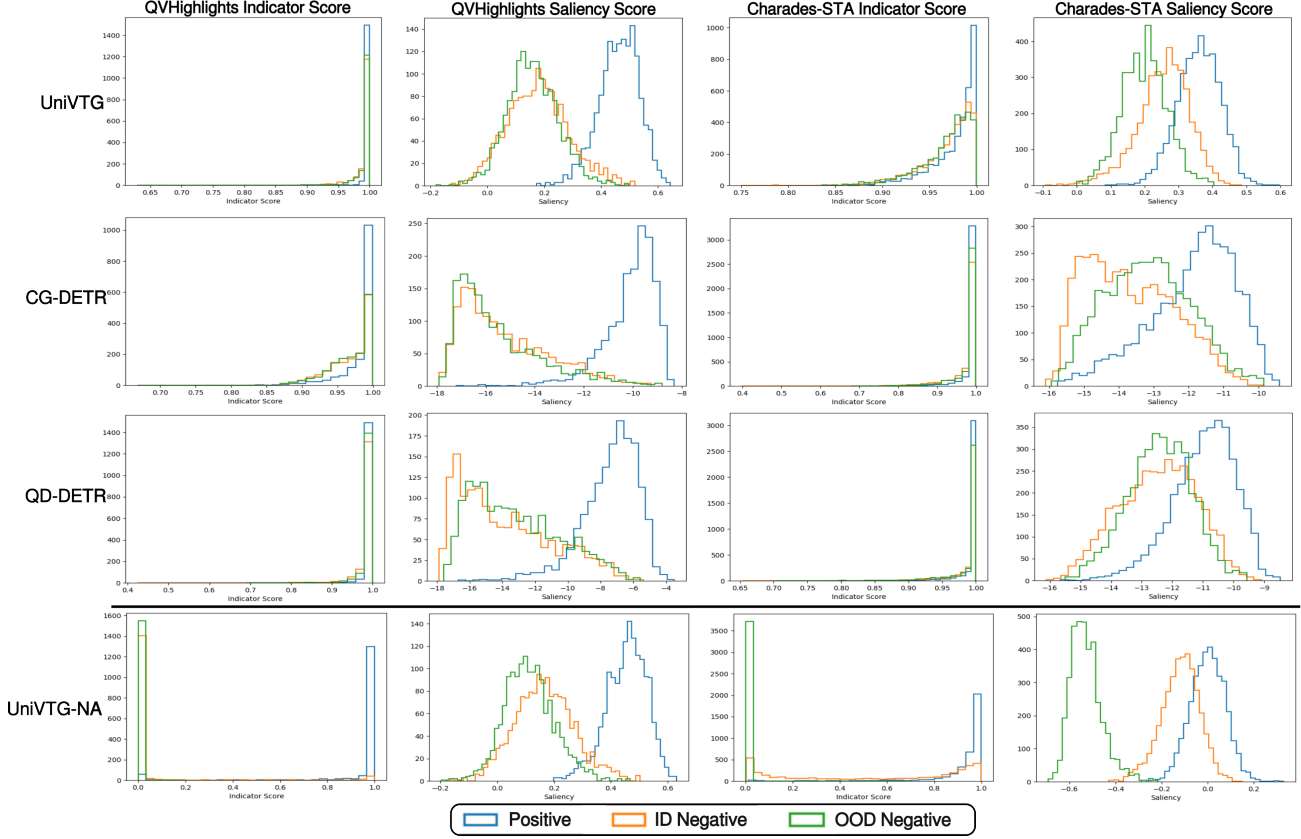


Figure 4. Histograms of indicator and saliency scores from UniVTG, CG-DETR and QD-DETR on the positive and negative queries. Bottom row: Indicator and saliency scores from UniVTG-NA.

| Method       | QVHighlights |        |                    |        | Charades-STA |        |                    |        |
|--------------|--------------|--------|--------------------|--------|--------------|--------|--------------------|--------|
|              | R1@0.5       | R1@0.7 | Rejection Acc. (%) |        | R1@0.5       | R1@0.7 | Rejection Acc. (%) |        |
|              |              |        | ID                 | OOD    |              |        | ID                 | OOD    |
| UniVTG [23]  | 67.35        | 52.65  | 0.00               | 0.00   | 60.22        | 38.55  | 0.00               | 0.00   |
| UniVTG-SVM   | 63.48        | 49.87  | 94.77              | 97.74  | 53.47        | 33.49  | 35.89              | 50.40  |
| UniVTG-NA    | 63.48        | 50.00  | 96.84              | 100.00 | 55.54        | 35.11  | 64.11              | 100.00 |
| CG-DETR [32] | 67.10        | 53.55  | 0.00               | 0.00   | 57.53        | 35.67  | 0.00               | 0.00   |
| CG-DETR SVM  | 62.84        | 50.26  | 95.55              | 95.41  | 43.79        | 28.44  | 82.9               | 74.52  |
| CG-DETR-NA   | 63.03        | 49.42  | 91.61              | 99.94  | 50.05        | 32.34  | 71.96              | 100.00 |
| QD-DETR [32] | 62.00        | 46.26  | 0.00               | 0.00   | 59.11        | 36.75  | 0.00               | 0.00   |
| QD-DETR SVM  | 48.26        | 37.42  | 96.32              | 95.09  | 46.67        | 30.05  | 76.91              | 81.59  |
| QD-DETR-NA   | 59.10        | 44.52  | 90.58              | 99.74  | 55.19        | 34.89  | 55.91              | 100.00 |

Table 2. Results of training using negative queries for the proposed Negative-Aware Video Moment Retrieval task. We compare using an SVM on top of UniVTG [23] with the proposed method UniVTG-NA across both QVHighlights and Charades-STA. Our proposed method loses less R1@ $\theta$  performance compared to using an SVM whilst improving upon rejection accuracy for both in-domain (ID) and out-of-domain (OOD) negatives. We also display corresponding results for CG-DETR-NA and QD-DETR-NA.

CG-DETR and QD-DETR, the Recall scores are generally much lower than the negative-aware methods. The reduction in Recall scores is largely due to false negative classification, where positive queries are incorrectly classed as

negative and therefore are not included in the Recall score. The negative-aware methods retain moment retrieval performance close to the original model, while the SVM methods typically suffer a much larger decrease. This is particularly

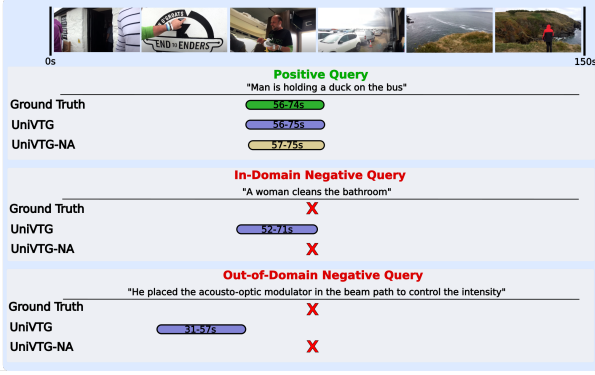


Figure 5. Qualitative result of UniVTG-NA on QVHighlights.

apparent for Charades-STA, where the lack of ground truth saliency scores makes it more difficult for the base models to learn to distinguish between positive and negative queries. The results show the particular importance of training with a classification head for datasets without ground truth saliency scores available. UniVTG-NA in particular improves on or matches the SVM across all metrics for both datasets, and shows the strongest overall performance in the combined moment retrieval and negative rejection tasks.

Overall, the performance on these datasets demonstrates the effectiveness of negative-aware training. Much of the video moment retrieval performance is retained while a large proportion of negative queries are rejected. While there is a trade-off between moment recall scores and negative rejection, this allows the model to be more robust to a wider variety of input queries and give more informative responses to those using the model. We now focus on UniVTG-NA for qualitative results and ablation.

**Qualitative Results.** Examples of predictions comparing UniVTG-NA and UniVTG are displayed in Figure 5. We see that UniVTG-NA retains accurate moment predictions while rejecting both ID and OOD negative queries, which UniVTG is not designed to do. Further qualitative results are found in the supp.

**Indicator and Saliency Scores for UniVTG-NA.** We inspect the indicator and saliency score outputs of the trained UniVTG-NA model in the bottom row of Figure 4, to compare with the equivalent scores from the video moment retrieval models shown in the upper rows. We can see that through our training scheme, the indicator scores become strong signifiers of whether the query is positive or negative, with the scores becoming clearly separable. The saliency scores are also strong indicators, with the Charades-STA saliency scores being much more separable, enabling a clearer distinction between positive and negative queries despite the lack of ground truth saliency scores. This shows that our added classification head and adjusted negative loss training scheme allows the classifier head to learn to classify positive/negative queries while also enabling the indicator

| UniVTG-NA |               | R1@0.5 | R1@0.7 | Rejection Accuracy (%) |               |
|-----------|---------------|--------|--------|------------------------|---------------|
| In-domain | Out-of-domain |        |        | In-Domain              | Out-of-Domain |
| ✓         | ✓             | 63.48  | 50.0   | 96.84                  | 100           |
| ✓         | -             | 61.03  | 47.87  | 96.80                  | 98.0          |
| -         | ✓             | 66.45  | 52.71  | 32.8                   | 100           |

Table 3. Effect of including in-domain negatives and out-of-domain negatives in training on the QVHighlights dataset.

and saliency heads to learn and reinforce this signal.

**Importance of In-Domain and Out-of-Domain.** We present results in Table 3 of model performance when trained with just ID negatives or just OOD negatives. The model trained with just ID negatives achieves weaker performance on rejecting both ID and OOD negatives, while R1@k scores also decrease due to an increased number of false negative predictions. When trained with just OOD negatives, the model is able to retain its 100% rejection of OOD negatives, but performance on ID negative rejection is much lower at  $\sim 33\%$ . The increased R1@k score is a by-product of the weakened rejection ability, as fewer false negative predictions are made. The benefit of fewer false negatives is far outweighed by the detriment of low negative rejection. These results indicate that ID and OOD negative queries offer a complementary training signal, mutually boosting performance.

**Limitations.** We note two limitations of the Negative-Aware approach: firstly, there remains a trade-off between localising positive moments and rejecting negative query sentences. Secondly, the pseudo-saliency scores used for datasets without ground truth human-annotated saliency scores are not as informative, which results in weaker negative rejection performance particularly in the in-domain case, as shown with the Charades-STA results.

## 5. Conclusion

In this work we have proposed the task of Negative-Aware Video Moment Retrieval, which incorporates negative query rejection into the standard video moment retrieval task. We have analysed the ability of current video moment retrieval methods to adapt to Negative-Aware Video Moment Retrieval and proposed negative-aware training which is specifically designed for this task. We have presented results for two new evaluation benchmarks on the QVHighlights and Charades-STA datasets. We have demonstrated the effectiveness of our method at rejecting negatives while maintaining high moment retrieval performance.

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